Parameter Selection for Full-Model Sensitivity Analysis

Fix vs. Sweep Recommendations for 896-Node SSWD-EvoEpi Model

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Abstract

We recommend **13 free (SWEEP) parameters** and 47 fixed parameters for the full-model (896-node) Morris + Sobol sensitivity analysis of the SSWD-EvoEpi agent-based model. This reduction from 60 total parameters makes the analysis computationally feasible (∼5 hours for Morris, ∼4 days for Sobol on a 48-core Xeon). Parameter selection is based on (1) SA R4 Sobol total-order indices from 25,088 model evaluations, (2) 134 calibration rounds on the full 896-node model, and (3) structural analysis of model changes since SA R4. Seven of the 13 SWEEP parameters are *new* (not in SA R4), reflecting the fundamental model restructuring for dynamic environmental pathogen pools, wavefront disease spread, and enclosedness-based flushing.

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1 Executive Summary

The SSWD-EvoEpi model has 60 parameters across 7 modules. The full-model SA on the 896-node spatial network requires ∼35 minutes per run, making exhaustive analysis of all 60 parameters infeasible. We reduce the free parameter count to 13.

Key Decisions

1. **13 SWEEP parameters**: 6 from SA R4 top tier (SA-validated) + 7 new parameters (untested, high expected influence).
2. **47 FIX parameters**: 41 from SA R4 bottom tiers ($S_T < 0.04$) + 6 new parameters with literature constraints or structural roles.
3. **P_env_max is structurally obsolete**: SA R4's #3 parameter ($S_T = 0.251$) is no longer used in the dynamic P_env model. Its role is absorbed by `alpha_env`, `delta_env`, and `P_env_floor`.
4. **alpha_env is expected to dominate**: In 134 calibration rounds, this single parameter drove more outcome variation than any other. It must be swept.

2 Background: SA R4 Results

SA R4 evaluated 47 parameters on an 11-node stepping-stone network. Table 1 shows the top 13 parameters by Sobol S_T for `pop_crash_pct`.

Table 1: SA R4 top 13 parameters by Sobol S_T (pop_crash_pct). Parameters above the dashed line are included in the SWEEP set; those below are fixed.

Rank	Parameter	Module	S_1	S_T	Morris μ^*	Decision
1	K_half	Disease	0.182	0.456	0.622	**SWEEP**
2	a_exposure	Disease	0.112	0.337	0.379	**SWEEP**
3	P_env_max	Disease	0.049	0.251	0.598	FIX (obsolete)
4	sigma_2_eff	Disease	0.041	0.232	0.431	**SWEEP**
5	sigma_D	Disease	0.075	0.141	0.211	FIX (borderline)
6	T_vbnc	Disease	−0.015	0.040	0.355	**SWEEP**
7	k_growth	Population	−0.009	0.035	0.633	**SWEEP**
8	peak_width_days	Spawning	0.018	0.035	0.391	FIX (borderline)
9	settler_survival	Population	0.010	0.033	0.509	**SWEEP**
10	target_mean_r	Genetics	0.008	0.021	0.236	FIX (borderline)
11	n_resistance	Genetics	0.029	0.020	0.525	**SWEEP**
12	sigma_1_eff	Disease	0.005	0.016	0.245	FIX
13	mu_I2D_ref	Disease	0.000	0.014	0.419	FIX

Bottom quartile: All 34 parameters ranked 14–47 have $S_T < 0.010$. These are unambiguously safe to fix.

3 Model Changes Since SA R4

Five structural changes affect parameter importance:

1. **Dynamic P_env** replaces static `P_env_max`. Environmental pathogen concentration is now driven by host shedding (`alpha_env`), natural decay (`delta_env`), and a community-maintained floor (`P_env_floor`).

2. **Wavefront disease spread** with cumulative dose threshold (`CDT`) and long-range dispersal (`wavefront_D_P`).
3. **Enclosedness-based flushing** via GIS-derived metrics, with nonlinearity exponent `n_connectivity`.
4. **Pathogen thermal adaptation**: per-site `T_vbnc_local` evolves toward local SST, bounded by `T_vbnc_min` = 9°C.
5. **Community virulence evolution**: per-site `v_local` evolves based on host density and temperature.

These changes introduce 13 new parameters not evaluated in SA R4. Seven of these are recommended for SWEEP.

4 SWEEP Parameters (13)

4.1 Disease Transmission Core (3)

Parameter	Default	Min	Max	Scale	Reason
K_half	87,000	20,000	200,000	log	(E) $S_T = 0.456$
a_exposure	0.75	0.30	1.50	linear	(E) $S_T = 0.337$
sigma_2_eff	50.0	10	250	log	(E) $S_T = 0.232$

These three parameters form the irreducible dose-response and shedding feedback kernel. SA R4 showed they collectively account for >70% of total-order variance for population crash. Their role is unchanged by the model restructuring.

4.2 Dynamic P_env System (3)

Parameter	Default	Min	Max	Scale	Reason
alpha_env	0.10	0.01	0.50	log	(F,G,H) Gradient driver
delta_env	0.05	0.01	0.20	log	(F,G,H) Pool persistence
P_env_floor	50.0	5	500	log	(F,G,H) Chronic pressure

`alpha_env` is **the single most important calibration parameter** discovered in rounds W01–W134. It controls the fraction of host shedding entering the environmental pathogen pool—the mechanism creating the north-south recovery gradient. `delta_env` determines pool persistence (half-life 3.5–70 days across the sweep range). `P_env_floor` sets chronic disease pressure from the multi-species vibrio community.

4.3 Wavefront Dynamics (2)

Parameter	Default	Min	Max	Scale	Reason
CDT	10.0	1	100	log	(F,H) Wavefront speed
wavefront_D_P	150.0	50	500	log	(F,H) Long-range dispersal

CDT and `wavefront_D_P` jointly control crash timing—the observed 3-year south-to-north wavefront (Gravem et al. 2021). Higher CDT means more accumulated dose before activation (slower wavefront); larger `wavefront_D_P` means further pathogen reach per timestep.

4.4 Spatial Structure (1)

Parameter	Default	Min	Max	Scale	Reason
n_connectivity	1.0	0.3	3.0	linear	(F,G,H) Flushing nonlinearity

Controls whether flushing rate drops sharply at moderate enclosedness ($n < 1$) or mostly at high enclosedness ($n > 1$). Flushing literature (Liu et al. 2019) suggests $n \approx 0.5$–0.7 but with substantial uncertainty. Directly modulates pathogen retention in enclosed sites (PWS, Hood Canal).

4.5 Temperature & Demography (4)

Parameter	Default	Min	Max	Scale	Reason
T_vbnc	12.0	8	15	linear	(E,H) $S_T = 0.040$
k_growth	0.08	0.03	0.15	linear	(E) $S_T = 0.035$
settler_survival	0.001	0.005	0.10	log	(E) $S_T = 0.033$
n_resistance	17	5	30	linear	(E,H) $S_T = 0.275^*$

*`n_resistance` $S_T = 0.020$ for `pop_crash_pct` but $S_T = 0.275$ for `resistance_shift_mean`. It is the dominant parameter for all evolutionary outcomes.

5 FIX Parameters (47)

5.1 Reason Codes

Code	Meaning
A	SA R4 showed insensitive (low μ^*, low S_T)
B	Well-constrained by literature
C	Structural/design choice
D	Computationally necessary

5.2 Complete Fixed Parameter Table

Parameter	Module	Value	Code	Notes
Disease (original)				
sigma_1_eff	Disease	5.0	A	$S_T = 0.016$
sigma_D	Disease	15.0	A*	$S_T = 0.141$ in R4; alpha_env subsumes role
rho_rec	Disease	0.05	A	$S_T = 0.004$; Morris #1 inflated by interactions
mu_EI1_ref	Disease	0.233	A,B	$S_T = 0.004$; Prentice 2025
mu_I1I2_ref	Disease	0.434	A,B	$S_T = 0.009$
mu_I2D_ref	Disease	0.563	A,B	$S_T = 0.014$; borderline
P_env_max	Disease	500.0	C	Obsolete: replaced by dynamic P_env
T_ref	Disease	20.0	A,B	$S_T = 0.004$; *V. pectenicida* optimum
s_min	Disease	10.0	A,B	$S_T = 0.004$; Vibrio salinity req.
susc. multiplier	Disease	2.0	A	$S_T = 0.003$; superseded by genetics
immunosupp. dur.	Disease	28	A	$S_T = 0.005$

Parameter	Module	Value	Code	Notes
min_susc. age	Disease	0	A	$S_T = 0.010$
Disease (new)				
k_vbnc	Disease	1.0	C	Standard sigmoid steepness
T_vbnc_min	Disease	9.0	B	Vibrio biophysical floor (see thermal review)
path. adapt. rate	Disease	0.001	B,G	Calibration-narrowed
v_adapt_rate	Disease	0.001	G	Secondary to alpha_env
K_cv	Population	0.0	C,D	Stochastic, not gradient driver
Spatial				
D_L	Spatial	400.0	A	$S_T = 0.003$
D_P	Spatial	15.0	A,B	Standard kernel
alpha_self_fjord	Spatial	0.30	A	$S_T = 0.004$
alpha_self_open	Spatial	0.10	A	$S_T = 0.003$
phi_open	Spatial	0.80	C	Boundary condition
phi_fjord	Spatial	0.03	C	Boundary condition
Population				
F0	Population	10^7	A	$S_T = 0.004$
gamma_fert	Population	4.5	A	$S_T = 0.004$
alpha_srs	Population	1.35	A	$S_T = 0.004$
senescence_age	Population	50.0	A	$S_T = 0.003$
L_min_repro	Population	400.0	A	$S_T = 0.002$
Genetics				
n_tolerance	Genetics	17	A	$S_T = 0.004$
target_mean_r	Genetics	0.15	A	$S_T = 0.021$; degenerate with n_resistance
target_mean_t	Genetics	0.10	A	$S_T = 0.004$
target_mean_c	Genetics	0.02	A	$S_T = 0.003$
tau_max	Genetics	0.85	A	$S_T = 0.005$
q_init_beta_a	Genetics	2.0	A	$S_T = 0.004$
q_init_beta_b	Genetics	8.0	A	$S_T = 0.003$
Spawning				
peak_width_days	Spawning	60.0	A*	$S_T = 0.035$; borderline
p_spont. female	Spawning	0.012	A	$S_T = 0.004$
p_spont. male	Spawning	0.0125	A	$S_T = 0.004$
fem. max_bouts	Spawning	2	A	$S_T = 0.004$
ind. f→m	Spawning	0.80	A	$S_T = 0.004$
ind. m→f	Spawning	0.60	A	$S_T = 0.004$
readiness_ind.	Spawning	0.50	A	$S_T = 0.005$
Pathogen Evolution				
alpha_kill	Path. Evol.	2.0	A,B	$S_T = 0.006$; Anderson-May
alpha_shed	Path. Evol.	1.5	A,B	$S_T = 0.005$
alpha_prog	Path. Evol.	1.0	A,B	$S_T = 0.004$
gamma_early	Path. Evol.	0.3	A	$S_T = 0.004$
sigma_v_mut.	Path. Evol.	0.02	A	$S_T = 0.005$
v_init	Path. Evol.	0.5	A,B	$S_T = 0.005$

*Borderline decisions discussed in §6.

6 Borderline Decisions

6.1 sigma_D (FIXED at 15.0)

SA R4 ranked `sigma_D` #5 ($S_T = 0.141$). This is well above the SWEEP threshold. However, with dynamic P_env, `alpha_env` controls the fraction of *all* shedding entering the environmental pool, including contributions from dead animals. `sigma_D` and `alpha_env` are partially degenerate—both increase pathogen loading. Sweeping both creates identifiability problems. We fix `sigma_D` and sweep `alpha_env` because the latter is the more mechanistically fundamental parameter in the current model.

6.2 peak_width_days (FIXED at 60.0)

SA R4 ranked `peak_width_days` #8 ($S_T = 0.035$). It primarily affects spawning synchrony and recruitment timing but does not directly control any of the six target biological questions. In calibration (W01–W134), varying `peak_width_days` had minimal impact on the recovery gradient. The literature supports 60±15 days (Animal Diversity Web: spawning "between March and July" with "main peak in May and June").

6.3 target_mean_r (FIXED at 0.15)

SA R4 ranked `target_mean_r` #10 ($S_T = 0.021$). It sets the initial population mean resistance. However, it is partially degenerate with `n_resistance`: both control the initial genetic variance and evolutionary potential. We sweep `n_resistance` (controls mechanistic architecture) and fix `target_mean_r` (calibrated value).

7 Biological Questions Coverage

Table 3: Coverage of biological questions by SWEEP parameters.

Biological Question	Key SWEEP Parameters	Coverage
Recovery gradient (AK 50% vs CA-S 0.1%)	alpha_env, delta_env, P_env_floor, n_connectivity, T_vbnc	Excellent
Crash timing (wavefront speed)	CDT, wavefront_D_P, K_half, a_exposure	Excellent
Evolutionary dynamics	n_resistance, k_growth, settler_survival	Good
Pathogen adaptation	T_vbnc, alpha_env, delta_env	Moderate
Reintroduction outcomes	settler_survival, alpha_env, K_half	Good
La Niña rebound	T_vbnc, alpha_env, delta_env, k_growth	Good

8 Computational Cost

Table 4: Computational cost estimates. Platform: Xeon W-3365 (48 cores), 35 min/run average.

Phase	**Total Runs**	**Serial Hours**	**Wall Time (48×)**
Morris ($r = 20$, $k = 13$)	280	163	∼5 hours
Sobol ($N = 256$, $k = 13$)	7,168	4,181	∼4 days
Total	**7,448**	**4,344**	**∼5 days**

Formulas used:
- Morris: $r \times (k + 1) = 20 \times 14 = 280$ runs
- Sobol (Saltelli): $N \times (2k + 2) = 256 \times 28 = 7{,}168$ runs

If $N = 256$ yields wide confidence intervals, increase to $N = 512$ (14,336 runs, ∼8 days).

9 Recommended SA Metrics

Table 5: Output metrics mapping to biological questions.

Metric	**Question**	**Definition**
recovery_gradient	Recovery	AK-PWS recovery − CA-S recovery
rmse_log_recovery	Recovery	Log-space RMSE against 8 regional targets
wavefront_mae	Crash timing	MAE of disease arrival vs. Gravem 2021
final_pop_frac	Crash severity	Final / peak total population
resistance_shift_mean	Evolution	$\Delta\bar{r}$ from year 0 to final
v_local_gradient	Pathogen adapt.	Mean v_{local}(AK) − mean v_{local}(CA)
reintro_persistence	Reintroduction	Pop at released site, 5 yr post-release / released
la_nina_rebound	La Niña	Pop year 9 / pop year 7

Acknowledgments

SA R4 analysis: 25,088 model evaluations on Intel Xeon W-3365. Calibration rounds W01–W134: 896-node network with satellite SST forcing. Vibrio thermal limits review informed T_vbnc_min constraint. Flushing residence time literature informed n_connectivity range.